COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE212-Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	9/24/2026	**DATE SUBMITTED:**	9/24/2026
NAME:	Quioyo, Angelo M.	**INSTRUCTOR:**	Engr. Jimlord Quejado

ACTIVITY NO. **Midterm Skills Exam**

1. PROCEDURE OUTPUT

Sample output (students are allowed to add more steps as needed.)
Task 1: Prepare the Server
Ubuntu:

```
quioyo3@WorkStation:~/Desktop$ sudo apt update
[sudo] password for quioyo3:
Hit:1 http://ph.archive.ubuntu.com/ubuntu jammy InRelease
Get:2 http://ph.archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Get:3 http://ph.archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]
Get:4 http://ph.archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages [3,8
55 kB]
Get:5 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
Get:6 http://security.ubuntu.com/ubuntu jammy-security/main i386 Packages [909 k
B]
Get:7 http://security.ubuntu.com/ubuntu jammy-security/main amd64 Packages [3,57
9 kB]
Get:8 http://security.ubuntu.com/ubuntu jammy-security/main Translation-en [501
kB]
Get:9 http://security.ubuntu.com/ubuntu jammy-security/main amd64 DEP-11 Metadat
a [74.3 kB]
Get:10 http://security.ubuntu.com/ubuntu jammy-security/restricted i386 Packages
 [51.7 kB]
Get:11 http://security.ubuntu.com/ubuntu jammy-security/restricted amd64 Package
s [6,390 kB]
Get:12 http://security.ubuntu.com/ubuntu jammy-security/restricted Translation-e
n [1,229 kB]
Get:13 http://security.ubuntu.com/ubuntu jammy-security/universe i386 Packages [
709 kB]
```

```
quioyo3@WorkStation:~/Desktop$ sudo apt upgrade -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following packages were automatically installed and are no longer required:
  libfwupd2 libfwupdplugin5 libgcab-1.0-0 libsmbios-c2 python3-resolvelib
Use 'sudo apt autoremove' to remove them.
Get more security updates through Ubuntu Pro with 'esm-apps' enabled:
  ansible python3-xmltodict
Learn more about Ubuntu Pro at https://ubuntu.com/pro
The following NEW packages will be installed:
  linux-headers-6.8.0-138-generic linux-hwe-6.8-headers-6.8.0-138
  linux-hwe-6.8-tools-6.8.0-138 linux-image-6.8.0-138-generic
  linux-modules-6.8.0-138-generic linux-modules-extra-6.8.0-138-generic
  linux-tools-6.8.0-138-generic
The following packages have been kept back:
  dnsmasq-base libaudit-common libaudit1
The following packages will be upgraded:
  bind9-dnsutils bind9-host bind9-libs bsdextrautils bsdutils bubblewrap bzip2
  coreutils cpio diffutils eject fdisk fwupd ghostscript ghostscript-x
  gir1.2-gst-plugins-base-1.0 gir1.2-nm-1.0 gir1.2-polkit-1.0 gir1.2-soup-2.4
  gstreamer1.0-alsa gstreamer1.0-gl gstreamer1.0-gtk3
  gstreamer1.0-plugins-base gstreamer1.0-plugins-base-apps
```

```
quioyo3@WorkStation:~$ sudo apt install -y wget tar curl
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
tar is already the newest version (1.34+dfsg-1ubuntu0.1.22.04.6).
tar set to manually installed.
wget is already the newest version (1.21.2-2ubuntu1.5).
wget set to manually installed.
The following packages were automatically installed and are no longer required:
  libfwupd2 libfwupdplugin5 libgcab-1.0-0 libsmbios-c2
  linux-headers-6.8.0-124-generic linux-hwe-6.8-headers-6.8.0-124
  linux-hwe-6.8-tools-6.8.0-124 linux-image-6.8.0-124-generic
  linux-modules-6.8.0-124-generic linux-modules-extra-6.8.0-124-generic
  linux-tools-6.8.0-124-generic python3-resolvelib
Use 'sudo apt autoremove' to remove them.
The following NEW packages will be installed:
  curl
0 upgraded, 1 newly installed, 0 to remove and 3 not upgraded.
Need to get 194 kB of archives.
After this operation, 456 kB of additional disk space will be used.
Get:1 http://ph.archive.ubuntu.com/ubuntu jammy-updates/main amd64 curl amd64 7.
81.0-1ubuntu1.27 [194 kB]
Fetched 194 kB in 1s (240 kB/s)
Selecting previously unselected package curl.
```

CentOS:

Task 2: Create the Prometheus User

```
quioyo3@WorkStation:~$ sudo useradd  --no-create-home --shell /bin/false prometh
eus
[sudo] password for quioyo3:
quioyo3@WorkStation:~$
```

```
quioyo3@WorkStation:~$ mkdir /etc/prometheus
mkdir: cannot create directory ‘/etc/prometheus’: File exists
quioyo3@WorkStation:~$ mkdir /var/lib/prometheus
mkdir: cannot create directory ‘/var/lib/prometheus’: File exists
quioyo3@WorkStation:~$
```

CentOS:

```
[qangelo@localhost ~]$ sudo useradd --no-create-home --shell /bin/false node_ex
porter
[qangelo@localhost ~]$ cd /tmp
```

```
[qangelo@localhost tmp]$ wget https://github.com/node_exporter/node_exporter/rel
eases/download/v3.14.0/node_exporter-3.14.0.linux-amd64.tar.gz
--2026-09-24 09:42:53--  https://github.com/node_exporter/node_exporter/releases
/download/v3.14.0/node_exporter-3.14.0.linux-amd64.tar.gz
Resolving github.com (github.com)... 20.205.243.166
Connecting to github.com (github.com)|20.205.243.166|:443... connected.
HTTP request sent, awaiting response... 404 Not Found
2026-09-24 09:42:54 ERROR 404: Not Found.
```

Task 3: Create Prometheus Directories

Ubuntu:

```
quioyo3@WorkStation:~$ sudo mkdir -p /etc/prometheus
quioyo3@WorkStation:~$ sudo mikdir -p /var/lib/prometheus
sudo: mikdir: command not found
quioyo3@WorkStation:~$ sudo mkdir -p /var/lib/prometheus
quioyo3@WorkStation:~$
```

```
quioyo3@WorkStation:~$ sudo chown -R prometheus:prometheus /etc/prometheus /var/
lib/prometheus
quioyo3@WorkStation:~$
```

```
quioyo3@WorkStation:~$ ls -ld /etc/prometheus /var/lib/prometheus
drwxr-xr-x 2 prometheus prometheus 4096 Sep 24 08:26 /etc/prometheus
drwxr-xr-x 2 prometheus prometheus 4096 Sep 24 08:02 /var/lib/prometheus
quioyo3@WorkStation:~$
```

Task 4: Download Prometheus

Ubuntu:

```
quioyo3@WorkStation:~$ cd /tmp
quioyo3@WorkStation:/tmp$ wget https://github.com/prometheus/prometheus/releases
/download/v3.14.0/prometheus-3.14.0.linux-amd64.tar.gz
--2026-09-24 08:09:50--  https://github.com/prometheus/prometheus/releases/downl
oad/v3.14.0/prometheus-3.14.0.linux-amd64.tar.gz
Resolving github.com (github.com)... 20.205.243.166
Connecting to github.com (github.com)|20.205.243.166|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://release-assets.githubusercontent.com/github-production-release
-asset/6838921/858be3cb-99c4-4d00-b806-a19381decb05?sp=r&sv=2018-11-09&sr=b&spr=
https&se=2026-09-24T01%3A06%3A28Z&rscd=attachment%3B+filename%3Dprometheus-3.14.
0.linux-amd64.tar.gz&rsct=application%2Foctet-stream&skoid=96c2d410-5711-43a1-ae
dd-ab1947aa7ab0&sktid=398a6654-997b-47e9-b12b-9515b896b4de&skt=2026-09-24T00%3A0
5%3A41Z&ske=2026-09-24T01%3A06%3A28Z&sks=b&skv=2018-11-09&sig=iIMV44DPV0X9hsOwYw
jGD%2BVHpOW81G6QVKt6D%2FkGnew%3D&jwt=eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.eyJpc3
MiOiJnaXRodWIuY29tIiwiYXVkIjoicmVsZWFzZS1hc3NldHMuZ2l0aHVidXNlcmNvbnRlbnQuY29tIi
wia2V5Ijoia2V5MSIsImV4cCI6MTc5MDIxMjE5MCwibmJmIjoxNzkwMjA4NTkwLCJwYXRoIjoicmVsZW
FzZWFzc2V0cHJvZHVjdGlvbi5ibG9iLmNvcmUud2luZG93cy5uZXQifQ.ckD6qlPDB5Bco-sa32Khiu4
joM-yTcudGFwZ7wXYCpE&response-content-disposition=attachment%3B%20filename%3Dpro
metheus-3.14.0.linux-amd64.tar.gz&response-content-type=application%2Foctet-strea
m [following]
--2026-09-24 08:09:50--  https://release-assets.githubusercontent.com/github-pro
duction-release-asset/6838921/858be3cb-99c4-4d00-b806-a19381decb05?sp=r&sv=2018-
11-09&sr=b&spr=https&se=2026-09-24T01%3A06%3A28Z&rscd=attachment%3B+filename%3Dp
rometheus-3.14.0.linux-amd64.tar.gz&rsct=application%2Foctet-stream&skoid=96c2d4
10-5711-43a1-aedd-ab1947aa7ab0&sktid=398a6654-997b-47e9-b12b-9515b896b4de&skt=20
26-09-24T00%3A05%3A41Z&ske=2026-09-24T01%3A06%3A28Z&sks=b&skv=2018-11-09&sig=iIM
V44DPV0X9hsOwYwjGD%2BVHpOW81G6QVKt6D%2FkGnew%3D&jwt=eyJ0eXAiOiJKV1QiLCJhbGciOiJI
UzI1NiJ9.eyJpc3MiOiJnaXRodWIuY29tIiwiYXVkIjoicmVsZWFzZS1hc3NldHMuZ2l0aHVidXNlcmN
vbnRlbnQuY29tIiwia2V5Ijoia2V5MSIsImV4cCI6MTc5MDIxMjE5MCwibmJmIjoxNzkwMjA4NTkwLCJ
wYXRoIjoicmVsZWFzZWFzc2V0cHJvZHVjdGlvbi5ibG9iLmNvcmUud2luZG93cy5uZXQifQ.ckD6qlPD
B5Bco-sa32Khiu4joM-yTcudGFwZ7wXYCpE&response-content-disposition=attachment%3B%2
0filename%3Dprometheus-3.14.0.linux-amd64.tar.gz&response-content-type=applicati
on%2Foctet-stream
Resolving release-assets.githubusercontent.com (release-assets.githubusercontent
.com)... 185.199.108.133, 185.199.110.133, 185.199.111.133, ...
Connecting to release-assets.githubusercontent.com (release-assets.githubusercon
tent.com)|185.199.108.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 107111714 (102M) [application/octet-stream]
Saving to: ‘prometheus-3.14.0.linux-amd64.tar.gz’

prometheus-3.14.0.l 100%[===================>] 102.15M  2.28MB/s    in 47s

2026-09-24 08:10:38 (2.17 MB/s) - ‘prometheus-3.14.0.linux-amd64.tar.gz’ saved [
107111714/107111714]
```

```
quioyo3@WorkStation:/tmp$ ls -l prometheus-3.14.0.linux-amd64.tar.gz
-rw-rw-r-- 1 quioyo3 quioyo3 107111714 Aug 18 15:36 prometheus-3.14.0.linux-amd6
4.tar.gz
quioyo3@WorkStation:/tmp$ tar -xvf prometheus-3.14.0.linux-amd64.tar.gz
prometheus-3.14.0.linux-amd64/
prometheus-3.14.0.linux-amd64/prometheus.yml
prometheus-3.14.0.linux-amd64/prometheus
prometheus-3.14.0.linux-amd64/NOTICE
prometheus-3.14.0.linux-amd64/LICENSE
prometheus-3.14.0.linux-amd64/promtool
quioyo3@WorkStation:/tmp$
```

]

Task 5: Install the Prometheus Binaries

```
quioyo3@WorkStation:/tmp$ sudo cp prometheus-3.14.0.linux-amd64/prometheus /usr/
local/bin && sudo cp prometheus-3.14.0.linux-amd64/promtool /usr/local/bin
quioyo3@WorkStation:/tmp$ sudo chown -R prometheus:prometheus /usr/local/bin/pro
metheus /usr/local/bin/promtool
quioyo3@WorkStation:/tmp$ sudo chown prometheus:prometheus /usr/local/bin/promet
heus /usr/local/bin/promtool
quioyo3@WorkStation:/tmp$
```

Task 6: Install the Prometheus Configuration Files

Ubuntu:

```
quioyo3@WorkStation:/tmp$ sudo cp prometheus-3.14.0.linux-amd64/prometheus.yml /
etc/prometheus
quioyo3@WorkStation:/tmp$
```

CentOS:

Task 7: Validate the Configuration

Ubuntu:

```
quioyo3@WorkStation:/tmp$ promtool check config /etc/prometheus/prometheus.yml
Checking /etc/prometheus/prometheus.yml
 SUCCESS: /etc/prometheus/prometheus.yml is valid prometheus config file syntax
```

CentOS:

Task 8: Create the systemd Service

Task 9: Start and Enable Prometheus

```
quioyo3@WorkStation:/tmp$ sudo systemctl daemon-reload
quioyo3@WorkStation:/tmp$ sudo systemctl restart prometheus
quioyo3@WorkStation:/tmp$ sudo systemctl status prometheus
● prometheus.service - Prometheus
     Loaded: loaded (/etc/systemd/system/prometheus.service; enabled; vendor preset: enabled)
     Active: active (running) since Thu 2026-09-24 09:51:36 CST; 5s ago
   Main PID: 18193 (prometheus)
      Tasks: 9 (limit: 8121)
     Memory: 30.6M
        CPU: 319ms
     CGroup: /system.slice/prometheus.service
             └─18193 /usr/local/bin/prometheus --config.file=/etc/prometheus/prometheus.yml --storage.tsdb.path=/var/lib/prome>

Sep 24 09:51:36 WorkStation prometheus[18193]: time=2026-09-24T09:51:36.218+08:00 level=INFO source=head.go:832 msg="Replaying>
Sep 24 09:51:36 WorkStation prometheus[18193]: time=2026-09-24T09:51:36.219+08:00 level=INFO source=head.go:927 msg="WAL segme>
Sep 24 09:51:36 WorkStation prometheus[18193]: time=2026-09-24T09:51:36.219+08:00 level=INFO source=head.go:964 msg="WAL repla>
Sep 24 09:51:36 WorkStation prometheus[18193]: time=2026-09-24T09:51:36.220+08:00 level=INFO source=main.go:1489 msg="filesyst>
Sep 24 09:51:36 WorkStation prometheus[18193]: time=2026-09-24T09:51:36.220+08:00 level=INFO source=main.go:1492 msg="TSDB sta>
Sep 24 09:51:36 WorkStation prometheus[18193]: time=2026-09-24T09:51:36.220+08:00 level=INFO source=main.go:1690 msg="Loading >
Sep 24 09:51:36 WorkStation prometheus[18193]: time=2026-09-24T09:51:36.220+08:00 level=INFO source=main.go:1106 msg="TSDB ret>
Sep 24 09:51:36 WorkStation prometheus[18193]: time=2026-09-24T09:51:36.271+08:00 level=INFO source=main.go:1729 msg="Complete>
Sep 24 09:51:36 WorkStation prometheus[18193]: time=2026-09-24T09:51:36.271+08:00 level=INFO source=main.go:1453 msg="Server i>
Sep 24 09:51:36 WorkStation prometheus[18193]: time=2026-09-24T09:51:36.271+08:00 level=INFO source=manager.go:211 msg="Starti>
lines 1-20/20 (END)
```

Task 10: Verify Prometheus

```
quioyo3@WorkStation:/tmp$ sudo ss -tulpn | grep 9090
tcp   LISTEN 0      4096               *:9090             *:*    users:(("prometheus",pid=18193,fd=6))
quioyo3@WorkStation:/tmp$ sudo netstat -tulpn | grep 9090
tcp6       0      0 :::9090                 :::*                    LISTEN      18193/prometheus
quioyo3@WorkStation:/tmp$ curl http://localhost:9090/-/healthy
Prometheus Server is Healthy.
```

Task 11: Access Prometheus Through a Browser

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

Lab activities, templates, and related instructional materials shall not be reuploaded, reproduced, distributed, or shared with external organizations or third parties without the prior express written permission of the Technological Institute of the Philippines.